

4.3 Primes

Definition

- An integer $p > 1$ is called a **prime number** if the only positive factors of p are 1 and p .

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, ...

4.3 Primes

Definition

- An integer $p > 1$ is called a **prime number** if the only positive factors of p are 1 and p .
- An integer $p > 1$ and is not prime is called **composite**.

Theorem

There are infinitely many primes.

Current record of largest known prime (October 12, 2024)

$$2^{136,279,841} - 1, \quad (41,024,320 \text{ digits}).$$

However, there are still infinitely many unknown primes larger than this one.

The Fundamental Theorem of Arithmetic

Every integer greater than 1 can be written uniquely as a prime or as a product of two or more primes (where the prime factors are written in the order of increasing size).

$$n = p_1^{k_1} p_2^{k_2} \cdots p_m^{k_m},$$

where

- $p_1, p_2, \dots, p_m$ are prime factors of n , and
- each $k_1, k_2, \dots, k_m$ are positive integers.

$$60 = 5 \times 12 = 5 \times 3 \times 4 = 5 \times 3 \times 2^2 = 2^2 \times 3 \times 5.$$

Definition: Greatest Common Divisor

Given integers a and b , not both zero.

- The largest integer d such that $d \mid a$ and $d \mid b$ is called the **greatest common divisor** of a and b , denoted by $\gcd(a, b)$.
- We say a and b are **relatively prime** or **coprime** if $\gcd(a, b) = 1$.

Example. Find the greatest common divisor of 24 and 36.

Factorize $24 = 2^3 \times 3^1$ and $36 = 2^2 \times 3^2$

For each prime, take the smaller power to form the GCD: $2^2 \times 3^1 = 12$.

Example. 63 and 128 are relatively prime.

$63 = 7 \times 9 = 7 \times 3^2$ has prime factors 3 and 7, and $128 = 2^7$ has a prime factor 2.

There is no common prime factors appearing in both factorizations of 63 and 128.

So $\gcd(63, 128) = 1$, which means they are relatively prime.

A useful Lemma

Let $a = bq + r$, where a, b, q , and r are integers. Then

- d divides a and b if and only if d divides b and r ;
- $\gcd(a, b) = \gcd(b, r)$.

The Euclidean Algorithm (for calculating GCD)

Given integers a and b with $a \geq b > 0$, let $r_0 = a$ and $r_1 = b$,

$$a = bq_1 + r_2 \qquad r_2 < b,$$

$$b = r_2q_2 + r_3 \qquad r_3 < r_2,$$

$$r_2 = r_3q_3 + r_4 \qquad r_4 < r_3,$$

$$\vdots$$

$$r_{n-2} = r_{n-1}q_{n-1} + r_n \qquad r_n < r_{n-1},$$

$$r_{n-1} = r_nq_n.$$

Then

$$\begin{aligned} \gcd(a, b) &= \gcd(b, r_2) = \gcd(r_2, r_3) = \cdots \\ &= \gcd(r_{n-1}, r_n) = r_n. \end{aligned}$$

Example. Find the greatest common divisor of 222 and 159.

$$\begin{array}{l} \boxed{222} = \boxed{159} \times 1 + \boxed{63} \\ \quad \swarrow \quad \searrow \\ 159 = 63 \times 2 + \boxed{33} \\ \quad \swarrow \quad \searrow \\ 63 = 33 \times 1 + \boxed{30} \\ \quad \swarrow \quad \searrow \\ 33 = 30 \times 1 + \boxed{3} \quad \text{gcd}(222, 159) = 3. \\ \quad \swarrow \quad \searrow \\ 30 = 3 \times 10 \end{array}$$

Extended Euclidean algorithm (the first part).

Bézout's identity

If a and b are positive integers, there exist integers s and t such that

$$\gcd(a, b) = sa + tb.$$

$3 = 222s + 159t$ for some integers s and t .

One of s and t is positive, and another negative.

Bézout's identity

If a and b are positive integers, there exist integers s and t such that

$$\gcd(a, b) = sa + tb.$$

Example. Express $\gcd(222, 159)$ as a linear combination of 222 and 159. *Extended Euclidean algorithm (the second part).*

$3 = \boxed{33} - 1 \times \boxed{30}$ write 3 as a linear comb of the previous two remainders

$= \boxed{159} - 2 \times \boxed{63} - 1 \times (\boxed{63} - \boxed{33})$ Continue to write each larger remainder
 $= 159 - 3 \times 63 + 33$ into a linear comb. of even larger remainders

Simplify and repeat this step

$$\begin{aligned} &= 159 - 3 \times (222 - 159) + (159 - 2 \times 63) \\ &= 5 \times 159 - 3 \times 222 - 2 \times 63 \\ &= 5 \times 159 - 3 \times 222 - 2 \times (222 - 159) \\ &= 7 \times 159 - 5 \times 222 \end{aligned}$$

$3 = \gcd(159, 222) = (-5) \times 222 + 7 \times 159$, so $s = -5$ and $t = 7$.

4.4 Solving Congruences

Division in modulo arithmetic

If $\gcd(c, m) = 1$, then

$$ac \equiv bc \pmod{m} \Rightarrow a \equiv b \pmod{m}.$$

Counter-example: Take $c = 2$ and $m = 6$,

$$2 \times 2 = 4 \equiv 4 + 6 = 10 = 2 \times 5 \pmod{6}$$

but $2 \not\equiv 5 \pmod{6}$ (because we divide a **zero-divisor!!**)

Division in modulo arithmetic, Version 2

If $ac \equiv bc \pmod{m}$, then

$$a \equiv b \pmod{\frac{m}{\gcd(c, m)}}$$

(because $\gcd(c, \frac{m}{\gcd(c, m)}) = 1$.)

Bezout identity: If $\gcd(a,b) = 1$, which means a and b are coprime, then there exists integers s and t such that $as + bt = 1$.

If we take mod a to $as + bt = 1$, we get $bt \equiv 1 \pmod{a}$, so $t \equiv \frac{1}{b} \pmod{a}$.
So t is invertible modulo a , the inverse of t is b modulo a .

Definition

- If $ab \equiv 1 \pmod{m}$, we say that a is **invertible** modulo m , and b is an **inverse** of a modulo m .
- In this case, we write b as $a^{-1} \pmod{m}$.

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$$a^{-1} = 1/a$$

Theorem

$$a \text{ is invertible modulo } m \iff \gcd(a, m) = 1.$$

From $3 = \gcd(222, 159)$, divide everything by 3:

$$1 = \gcd(74, 53) = (-5) \times 74 + 7 \times 53$$

$$\text{mod } 74: 1 = 7 \times 53 \pmod{74}, \text{ so } 53^{-1} = 7 \pmod{74}$$

$$\begin{aligned} \text{mod } 53: 1 &= (-5) \times 74 \pmod{53}, \text{ so } 74^{-1} = -5 = 48 \pmod{53} \\ 74 \times 48 &= 3,552 = 1 \pmod{53} \end{aligned}$$

To compute the inverse of a mod m , either:

- by trial-and-error if m is small, or
- use Euclidean algorithm and Bezout's identity.

extended Euclidean algorithm

To compute the inverse of $a \bmod m$, either:

- by trial-and-error if m is small, or
- use Euclidean algorithm and Bezout's identity.

Example. Find an inverse of 3 modulo 7.

Trial-and-error:

$$\begin{array}{l} 3 \times 1 = 3 \\ 3 \times 2 = 6 \\ 3 \times 3 = 9 = 2 \\ 3 \times 4 = 12 = 5 \\ 3 \times 5 = 15 = 1 \\ 3 \times 6 = 18 \end{array} \quad \begin{array}{l} \\ \\ \text{mod } 7 \\ \\ \end{array}$$

$3^{-1} = 5 \bmod 7$



To compute the inverse of a mod m , either:

- by trial-and-error if m is small, or
- use Euclidean algorithm and Bezout's identity.

Example. Find an inverse of 3 modulo 7.

Example. Find an inverse of 21 modulo 37.

Use extended Euclidean algorithm to compute 1 as a linear comb. of a and b .

$$37 = 21 \times 1 + 16$$

$$21 = 16 \times 1 + 5$$

$$16 = 5 \times 3 + 1$$

$$1 = 16 - 3 \times 5$$

$$= (37 - 21) - 3 \times (21 - 16)$$

$$= 37 - 4 \times 21 + 3 \times 16$$

$$= 37 - 4 \times 21 + 3 \times (37 - 21)$$

$$= 4 \times 37 - 7 \times 21$$

The last result implies $1 = -7 \times 21 \bmod 37$, and so $21^{-1} = -7 = 30 \bmod 37$.

Theorem: Division in modulo arithmetic, Version 3

If $\gcd(a, m) = 1$, the linear congruence

$$ax \equiv b \pmod{m}$$

has a unique solution, which is $x = a^{-1}b \pmod{m}$.

Example. Solve the linear congruence

$$21x \equiv 8 \pmod{37}$$

$$x = \frac{8}{21} = 8 \times 21^{-1} \pmod{37}$$

From the last page: $1 = 4 \times 37 - 7 \times 21$

take mod 37: $1 = -7 \times 21 \pmod{37}$, so $21^{-1} = -7 \pmod{37}$

$$x = 8(-7) = -56 = \underline{18 \pmod{37}}.$$

check the answer: $21 \times 18 = 378 = 8 + 370 = 8 \pmod{37}$.

Definition

The integers $a_1, a_2, \dots, a_n$ are **pairwise relatively prime** or **pairwisely coprime** if

$$\gcd(a_i, a_j) = 1 \quad \text{for all } i \neq j.$$

(This is stronger than saying $\gcd(a_1, \dots, a_n) = 1$.)

Example. Determine whether 10, 21, and 22 are pairwise relatively prime.

$$\gcd(10, 21) = 1, \gcd(21, 22) = 1, \gcd(10, 22) = 2$$

10, 21, 22 are NOT pairwisely coprime



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Example. Determine whether 10, 21, and 22 are pairwise relatively prime.

Example. Determine whether 44, 63, and 65 are pairwise relatively prime.

Chinese Remainder Theorem (CRT)

Let

- $m_1, m_2, \dots, m_k$ be pairwisely coprime integers > 1 , and
- $a_1, a_2, \dots, a_k$ be arbitrary integers.

Then the system of k congruence equations

$$x \equiv a_1 \pmod{m_1}, \quad \dots, \quad x \equiv a_k \pmod{m_k}$$

has a unique solution modulo $m_1 m_2 \dots m_k$.

The solution formula of CRT

Put $M = m_1 m_2 \dots m_k$, and for each $i = 1, \dots, k$, let

y_i is the inverse of $M/m_i \pmod{m_i}$,

then the solution is (given by interpolation):

$$x = a_1(M/m_1)y_1 + \dots + a_k(M/m_k)y_k \pmod{M}.$$



Example. Solve the following system of congruences:

$$x \equiv \underline{1} \pmod{3}, \quad x \equiv \textcircled{2} \pmod{5}, \quad x \equiv \boxed{6} \pmod{7}.$$

Since the moduli 3, 5, 7 are pairwise coprime,
the solution in mod $(3 \times 5 \times 7 = 105)$ is unique.

For $m_1 = 3$, solve for the inverse of $\underline{5 \times 7 = 35} = 2 \pmod{3}$, which is $y_1 = \underline{2} \pmod{3}$

For $m_2 = 5$, solve for the inverse of $\textcircled{3 \times 7 = 21} = 1 \pmod{5}$, which is $y_2 = \textcircled{1} \pmod{5}$.

For $m_3 = 7$, solve for the inverse of $\boxed{3 \times 5 = 15} = 1 \pmod{7}$, which is $y_3 = \boxed{1} \pmod{7}$.

$$\begin{aligned} \text{The solution is } x &= \underline{1 \times (5 \times 7) \times 2} + \textcircled{2 \times (3 \times 7) \times 1} + \boxed{6 \times (3 \times 5) \times 1} \pmod{105} \\ &= 70 + 42 + 90 = 202 = -8 = 97 \pmod{105} \end{aligned}$$

Check the answer: $97 = 1 \pmod{3}$? Yes

$97 = 2 \pmod{5}$? Yes

$97 = 6 \pmod{7}$? Yes